Optimization: Critical Points & Lagrange Multipliers

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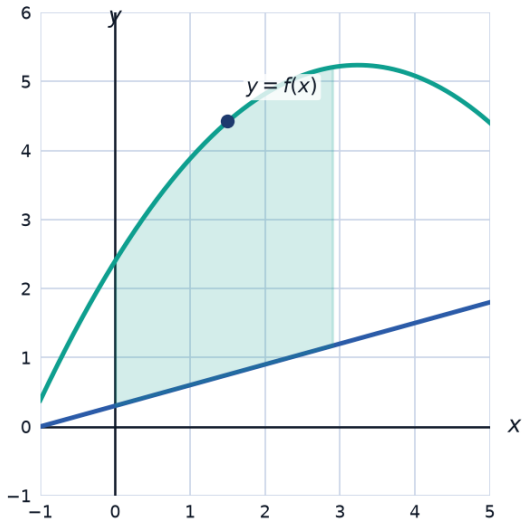
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find all critical points of the function $f(x, y) = x^3 + y^3 - 3xy$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Compute the scalar line integral with respect to arc length $\int_C (x + 2y + 3z)\, ds$, where C is the straight line segment from the origin $(0, 0, 0)$ to $(1, 2, 2)$.

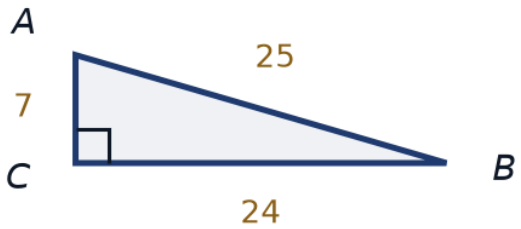


3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the volume of the solid inside the sphere $\rho = 4$ and above the cone $\phi = \frac{\pi}{4}$.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use Green's theorem to evaluate $\oint_C -y^3\, dx + x^3\, dy$, where C is the triangle with vertices $(0,\ 0)$, $(3,\ 0)$, $(0,\ 3)$, oriented counterclockwise.



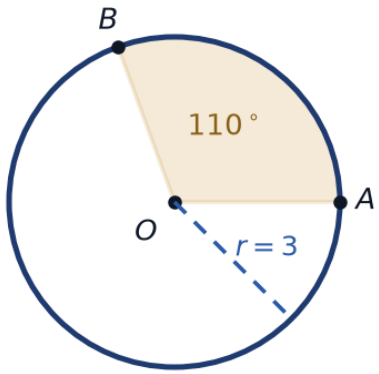
5 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For $f(x, y) = x^2 - 6xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Use the method of Lagrange multipliers to find the maximum and minimum values of $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 25$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find and classify all critical points of $f(x, y) = 2x^3 + xy^2 + 5x^2 + y^2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Multivariable Optimization: Critical Points, Hessians & Lagrange Multipliers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

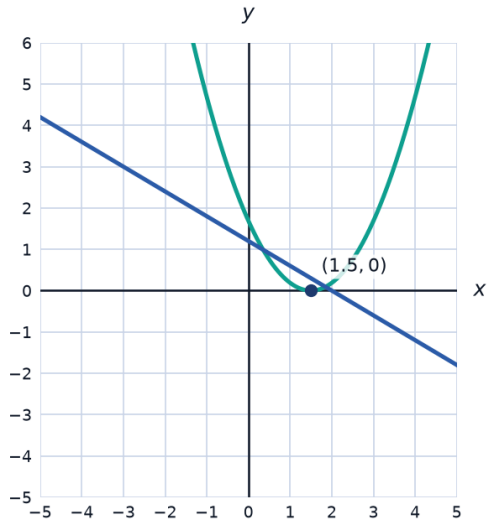
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Practice Problem 2: Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = \langle -y, 2x, (x + y)e^z \rangle$ and C is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = y + 1$, oriented counterclockwise when viewed from above.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Practice Problem 1: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle e^y, xe^y + (z+1)e^z, (z+2)e^z \rangle$ along the curve $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ for $0 \le t \le 1$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Practice Problem 1: Use the Divergence Theorem to calculate the outward flux of $\mathbf{F} = \langle x^3 + \sin(yz),\ y^3 + \cos(xz),\ z^3 + e^{xy} \rangle$ across the sphere $x^2 + y^2 + z^2 = 4$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Practice Problem 1: Use Green's Theorem to evaluate $\oint_C (y^3 - \ln(x^2+1))\,dx + (x^3 + e^{y^2})\,dy$, where C is the circle $x^2 + y^2 = 4$ oriented counterclockwise.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Practice Problem 2: Evaluate the line integral $\int_C (2y + \cos x)\,dx + (5x + \sin y)\,dy$ along the upper half of the unit circle $x^2 + y^2 = 1$ from $(1, 0)$ to $(-1, 0)$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the integral $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$ by reversing the order of integration.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

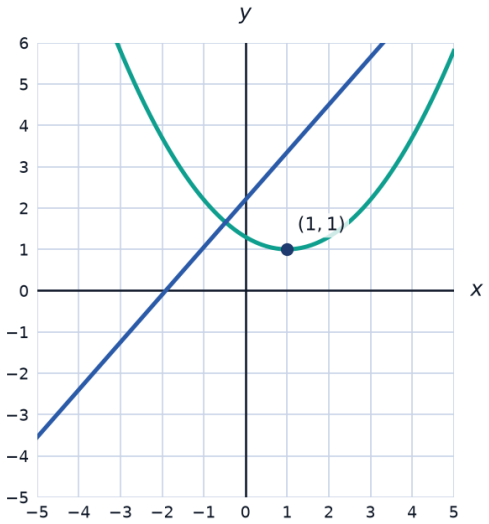
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Practice Problem 2: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = abla(x^3y^2 + \sin(xz))$ and C is the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \le t \le 2\pi$.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Double Integrals

Evaluate $\iint_D (x^2 + y^2)\,dA$, where D is the upper half-disk defined by $x^2 + y^2 \le 9$ with $y \ge 0$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Practice Problem 2: Find the mass of a solid right circular cylinder of radius R and height h whose density at any point is proportional to its distance from the axis of symmetry, $\delta(r, \theta, z) = kr$ for constant $k > 0$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Practice Problem 1: Evaluate $\iiint_E (x^2 + y^2 + z^2)\,dV$, where E is the solid ball $x^2 + y^2 + z^2 \le a^2$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Double Integrals

[Double Integrals, Fubini's Theorem, Polar Coordinates & The Change-of-Variables Jacobian · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

[Green's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What condition defines a critical point (x_0, y_0) of a differentiable function $f(x, y)$?

a. $\nabla f(x_0, y_0) = \mathbf{0}$, meaning both $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$

b. $f_{xx}(x_0, y_0) = 0$

c. $f(x_0, y_0) = 0$

d. $f_x(x_0, y_0) = f_y(x_0, y_0)$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the critical points of $f(x, y) = x^2 + y^2 - 4x + 6y + 10$.

a. $(2, -3)$

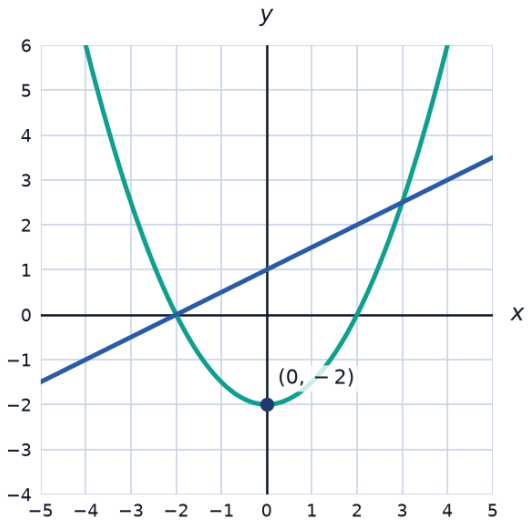
b. $(-2, 3)$

c. $(4, -6)$

d. $(0, 0)$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find all critical points of $g(x, y) = x^3 - 3x + y^2$.



a. $(1, 0)$ and $(-1, 0)$

b. $(1, 0)$ only

c. $(0, 0)$

d. $(3, 0)$

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the definition of the Hessian discriminant D in the Second Partials Test?

a. $D = f_{xx}f_{yy} - (f_{xy})^2$

b. $D = f_{xx} + f_{yy}$

c. $D = (f_{xy})^2 - f_{xx}f_{yy}$

d. $D = f_{xx}f_{yy} - f_{xy}$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If at a critical point $D > 0$ and $f_{xx} > 0$, what does the test conclude?

a. f has a local minimum at the point

b. f has a local maximum at the point

c. The point is a saddle point

d. The test is inconclusive

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Classify the critical point $(0, 0)$ for $f(x, y) = x^2 - y^2$.

a. Saddle point

b. Local minimum

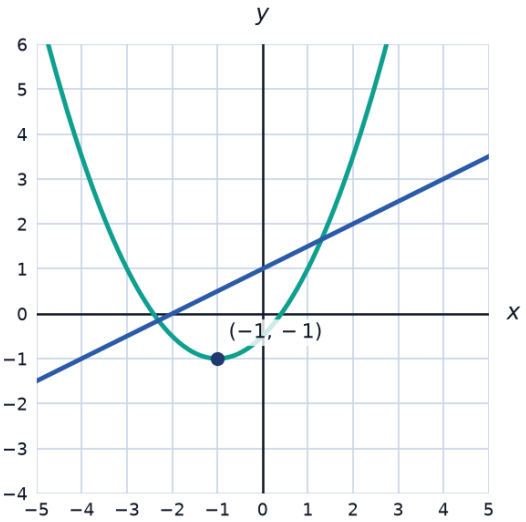
c. Local maximum

d. Inconclusive

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What does the Second Partials Test conclude when $D = 0$?



a. The test is inconclusive; higher-order analysis or algebraic examination is needed

b. The point is definitely a saddle point

c. The point is an absolute extremum

d. The function is not differentiable

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What vector equation governs constrained optimization of $f(x, y)$ subject to constraint $g(x, y) = k$?



a. $\nabla f = \lambda \nabla g$ (and $g(x, y) = k$)

b. $\nabla f \cdot \nabla g = 0$

c. $\nabla f + \nabla g = \mathbf{0}$

d. $\nabla f = \mathbf{0}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Maximize $f(x, y) = xy$ subject to the constraint $x + y = 10$ for $x, y > 0$.

a. Maximum is 25 at $(5, 5)$

b. Maximum is 24 at $(4, 6)$

c. Maximum is 21 at $(3, 7)$

d. Maximum is 16 at $(2, 8)$

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30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

What does the Lagrange multiplier λ represent physically/economically in constrained optimization?

a. The sensitivity / shadow price: the rate of change of the optimal value with respect to the constraint constant k ($\frac{df^*}{dk}$)

b. The curvature of the constraint surface

c. The distance from the origin to the optimal point

d. The second derivative along the constraint

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

For $f(x, y) = x^3 - 12x + y^2$, classify the critical point at $(-2, 0)$.

a. Saddle point

b. Local minimum

c. Local maximum

d. Inconclusive (D = 0)

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Find the x-coordinate of the critical point of $f(x, y) = x^2 - 6x + y^2 - 4y$.

a. 3

b. -6

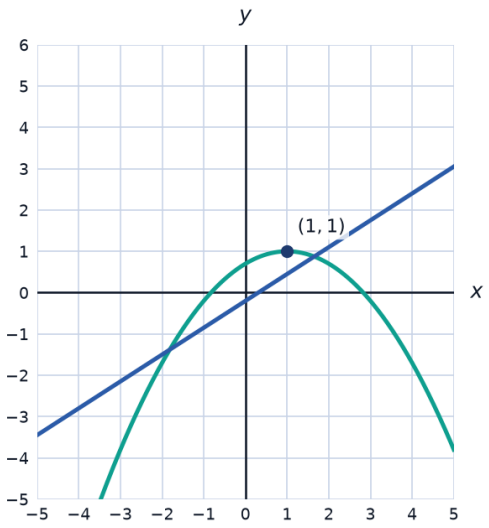
c. -3

d. 4

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Implicit function theorem for optimization: constrained critical points satisfy $abla_{\mathbf{x}} L = 0$ where $L = f - \lambda g$. Name of λ:



a. Lagrange multiplier

b. eigenvalue

c. gradient

d. curvature

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

To find candidates for the global max/min of f on $[a, b]$, you check critical points and:

a. the endpoints a, b

b. inflection points only

c. only $x = 0$

d. points where f is negative

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The method of Lagrange multipliers finds constrained extrema using multiplier λ. The critical value of λ itself:

a. is the rate of change of the optimal value w.r.t. the constraint constant

b. gives the extremum value

c. is always zero

d. equals $|ablaf|$

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Optimization with constraint (Lagrange): extremise f s.t. $g = 0$ via $ablaf = \lambda ablag$. Number of equations for n vars, 1 constraint:

a. $n + 1$

b. n

c. $n - 1$

d. $2n$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use spherical coordinates to find the volume of the upper half of the ball $x^2 + y^2 + z^2 \le 4$.

a. $\frac{16\pi}{3}$

b. $\frac{32\pi}{3}$

c. $\frac{8\pi}{3}$

d. 4π

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Use spherical coordinates to find the volume of the first-octant part of the ball $x^2 + y^2 + z^2 \le 16$.

a. $\frac{256\pi}{3}$

b. $\frac{64\pi}{3}$

c. $\frac{32\pi}{3}$

d. $\frac{16\pi}{3}$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Double Integrals

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle -y, x \rangle$ along the circle $x = 2\cos(t),\ y = 2\sin(t), 0 \le t \le \pi$.

a. -4π

b. 4π

c. 8π

d. 5π

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y,\ -x, 1 \rangle$ along the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle, 0 \le t \le 2\pi$.

a. π

b. 1

c. 0

d. -1

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the sphere $x^2 + y^2 + z^2 = 9$.

a. $\frac{2916\pi}{5}$

b. $\frac{5832\pi}{5}$

c. $\frac{1458\pi}{5}$

d. $-\frac{2916\pi}{5}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Double Integrals

Use Green's theorem to evaluate $\oint_C -y^3\,dx + x^3\,dy$, where C is the circle $x^2 + y^2 = 4$, oriented counterclockwise.

a. -24π

b. 48π

c. 12π

d. 24π

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Use Green's theorem to evaluate $\oint_C y\,dx + 3x\,dy$, where C is the boundary of the rectangle $[0, 1] \times [0, 1]$, oriented counterclockwise.

a. -2

b. 4

c. 1

d. 2

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Evaluate $\int_C x^2\,ds$ where C is the segment from $(0,\ 0)$ to $(5,\ 12)$.

a. $\frac{25}{3}$

b. $\frac{4225}{3}$

c. $\frac{328}{3}$

d. $\frac{325}{3}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Double Integrals

Use Green's theorem to find the area enclosed by the astroid $x = 1\cos^3 t,\ y = 1\sin^3 t$.

a. $\frac{3\pi}{8}$

b. π

c. $\frac{3\pi}{4}$

d. $\frac{3\pi}{16}$

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